

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I T.Ganesh(192425246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

T.Ganesh

Signature of the student

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1.5	1	10
Q2	2.5	3	2	1	1	9.5
Q3	2.5	3	2	1	0.5	9
Q4	2.5	3	2	1	0.5	9

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)

Wireshark packet capture showing a TCP connection setup. The packet list displays the following packets:

No.	Time	Source	Destination	Protocol	Length	Info
1311	33.958036	23.36.4.104	172.23.20.190	TLSv1.2	98	Server Key Exchange, Server Hello Done
1312	33.959823	172.23.20.190	23.36.4.104	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1313	34.013253	23.36.4.104	172.23.20.190	TCP	54	443 → 54019 [ACK] Seq=5601 Ack=291 Win=64128 Len=0
1314	34.041875	23.36.4.104	172.23.20.190	TLSv1.2	344	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
1315	34.086043	172.23.20.190	23.36.4.104	TLSv1.2	373	Application Data
1316	34.086348	172.23.20.190	23.36.4.104	TLSv1.2	288	Application Data
1349	34.108458	23.36.4.104	172.23.20.190	TCP	54	443 → 54019 [ACK] Seq=5891 Ack=610 Win=64128 Len=0
1350	34.108458	23.36.4.104	172.23.20.190	TCP	54	443 → 54019 [ACK] Seq=5891 Ack=844 Win=64128 Len=0
1351	34.111684	23.36.4.104	172.23.20.190	TLSv1.2	749	Application Data
1352	34.161132	172.23.20.190	23.36.4.104	TCP	54	54019 → 443 [ACK] Seq=844 Ack=6586 Win=64256 Len=0
1353	34.275442	172.23.20.190	23.36.4.104	TCP	54	54019 → 443 [RST, ACK] Seq=844 Ack=6586 Win=0 Len=0
77365	1805.422497	172.23.20.190	23.36.4.104	TCP	66	49674 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1

The packet details for the selected packet (1353) are shown below:

Frame 1353: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{FC0E9385-3790-43A7-B08E-3B30A9F8FB00}, id 0

Ethernet II, Src: 5c:b4:7e:e6:ae:f1 (5c:b4:7e:e6:ae:f1), Dst: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.20.190, Dst: 23.36.4.104

Transmission Control Protocol, Src Port: 54019, Dst Port: 443, Seq: 844, Ack: 6586, Len: 0

Source Port: 54019
Destination Port: 443
[Stream index: 5]
[Conversation completeness: Complete, WITH_DATA (47)]
[TCP Segment Len: 0]
Sequence Number: 844 (relative sequence number)
Sequence Number (raw): 2161407699
[Next Sequence Number: 844 (relative sequence number)]
Acknowledgment Number: 6586 (relative ack number)

The packet bytes are displayed at the bottom:

```
0000 24 d5 e4 6d bc f7 5c b4 7e e6 ae f1 08 00 45 00  $.m..\.~....E-
0010 00 28 5d cc 40 00 80 06 00 00 ac 17 14 be 17 24  .[]@... ..$
0020 04 68 d3 03 01 bb 80 d4 76 d3 e9 74 2f 95 50 14  -h.....v..t/P.
0030 00 00 dc 7b 00 00                                ...{..
```

2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)

Wireshark · Packet 24772 · Wi-Fi

- > Frame 24772: 255 bytes on wire (2040 bits), 255 bytes captured (2040 bits) on interface \Device\NPF_{FC0E9385-3790-43A7-B6BE-3B3DA9FBFB8D}, id 0
- > Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: 5c:b4:7e:e6:ae:f1 (5c:b4:7e:e6:ae:f1)
- > Internet Protocol Version 4, Src: 172.21.100.133, Dst: 172.23.20.190
- ▼ User Datagram Protocol, Src Port: 53, Dst Port: 56853
 - Source Port: 53
 - Destination Port: 56853
 - Length: 221
 - Checksum: 0x1375 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 3491]
 - > [Timestamps]
 - UDP payload (213 bytes)
- > Domain Name System (response)

0000	5c b4 7e e6 ae f1 24 d5 e4 6d bc f7 08 00 45 00	\.~.\$. m...E.
0010	00 f1 05 9e 00 00 7e 11 64 ee ac 15 64 85 ac 17	d...d...
0020	14 be 00 35 de 15 00 dd 13 75 75 36 81 80 00 01	...5....uu6....
0030	00 06 00 00 00 00 03 65 63 73 06 6f 66 66 69 63	e cs-offic
0040	65 03 63 6f 6d 00 00 01 00 01 c0 0c 00 05 00 01	e.com.....
0050	00 00 00 7a 00 1f 03 65 63 73 06 6f 66 66 69 63	...z...e cs-offic
0060	65 0e 74 72 61 66 66 69 63 6d 61 6e 61 67 65 72	e-trafficmanager
0070	03 6e 65 74 00 c0 2c 00 05 00 01 00 00 00 1e 00	.net.,,
0080	22 12 64 75 61 6c 2d 73 2d 30 30 30 35 2d 6f 66	".dual-s -0005-of
0090	66 69 63 65 06 63 6f 6e 66 69 67 05 73 6b 79 70	fice.con fig.skyp
00a0	65 c0 17 c0 57 00 05 00 01 00 00 23 fb 00 22 0a	e...W...#.."
00b0	65 63 73 2d 6f 66 66 69 63 65 06 73 2d 30 30 30	ecs-offi ce-s-000
00c0	35 0d 64 75 61 6c 2d 73 2d 6d 73 65 64 67 65 c0	5.dual-s -msedge.
00d0	46 c0 85 00 05 00 01 00 00 00 a4 00 02 c0 90 c0	F.....
00e0	b3 00 01 00 01 00 00 00 a6 00 04 34 7b 81 0e c0	4{...
00f0	b3 00 01 00 01 00 00 00 a6 00 04 34 7b 80 0e	4{..

3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)

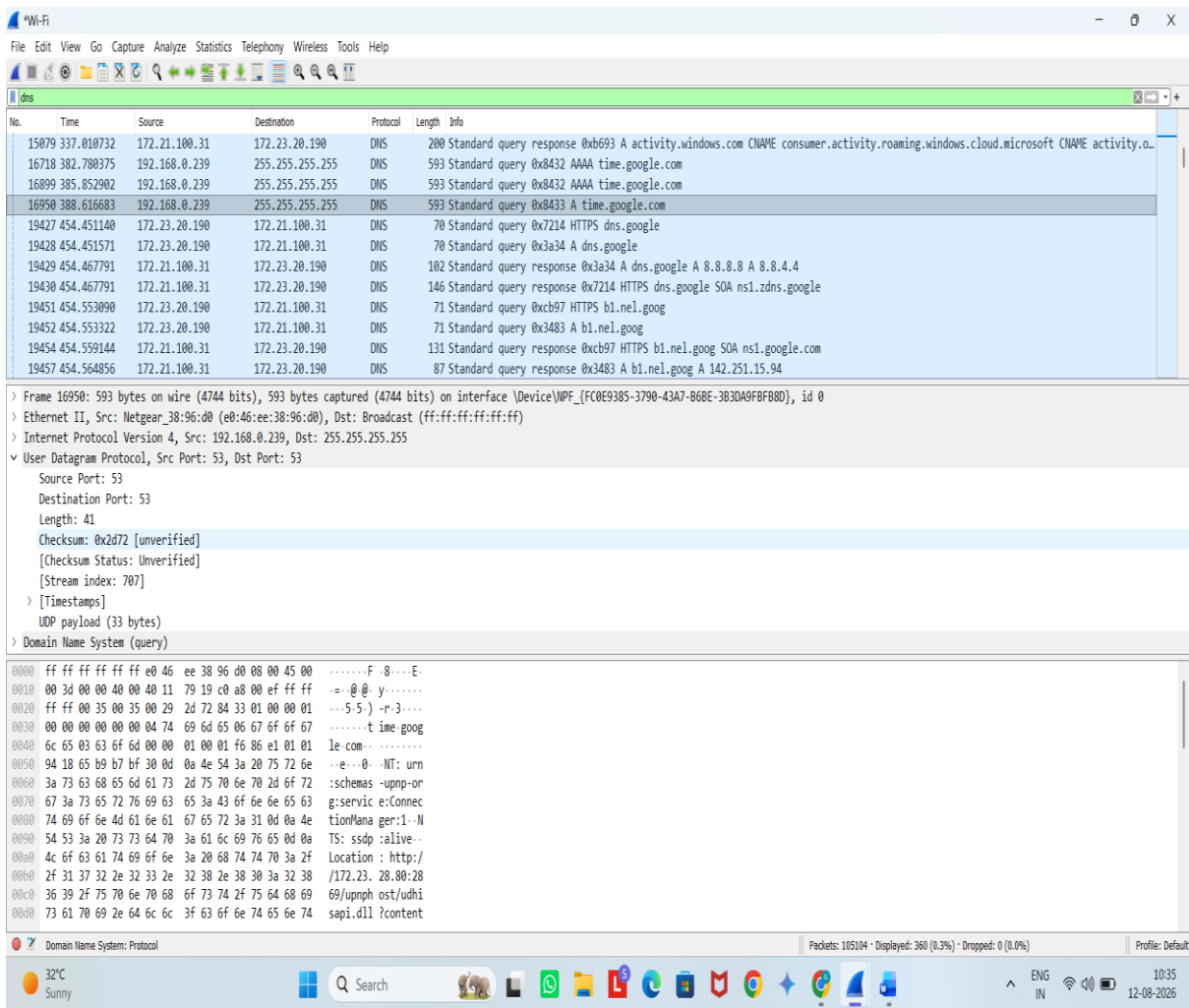
Wireshark · Packet 24898 · Wi-Fi

- > Frame 24898: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{FC0E9385-3790-43A7-B6BE-3B3DA9FBFB8D}, id 0
- > Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: 5c:b4:7e:e6:ae:f1 (5c:b4:7e:e6:ae:f1)
- > Internet Protocol Version 4, Src: 40.126.17.133, Dst: 172.23.20.190
- ▼ Transmission Control Protocol, Src Port: 443, Dst Port: 57242, Seq: 16986, Ack: 7639, Len: 1460
 - Source Port: 443
 - Destination Port: 57242
 - [Stream index: 25]
 - [Conversation completeness: Complete, WITH_DATA (47)]
 - [TCP Segment Len: 1460]
 - Sequence Number: 16986 (relative sequence number)
 - Sequence Number (raw): 82112552
 - [Next Sequence Number: 18446 (relative sequence number)]
 - Acknowledgment Number: 7639 (relative ack number)
 - Acknowledgment number (raw): 3576897122
 - 0101 = Header Length: 20 bytes (5)

0020	14	be	01	bb	df	9a	04	e4	f0	28	d5	33	1e	62	50	10		.(3-bP.
0030	40	01	a9	60	00	00	67	fd	59	0c	4f	91	78	58	15	c2	g...g	Y-0-xX..
0040	bf	f9	7d	5c	7f	12	a9	40	13	d1	6e	86	73	e3	f3	cc	..}\...@	..n-s...
0050	cd	c8	b4	ab	6c	67	00	b7	8f	14	da	38	07	e0	67	b9	...lg...	...8..g.
0060	fc	31	2b	27	57	cb	82	0d	d4	83	f6	df	4f	d5	17	a7	..1+'W...	0...
0070	d7	a1	4a	4c	c9	72	1e	af	73	ad	b2	38	17	2c	27	97	..JL...r...	s-8,',.
0080	a1	5b	6b	ed	c4	7c	b7	a8	6e	d6	f0	88	4c	7c	8f	e0	.[k... ...	n...L ..
0090	a8	aa	42	e4	5a	cf	62	a9	70	8b	77	03	4b	c2	9f	a7	..B-Z-b...	p-w-K...
00a0	1c	e5	5c	98	a8	00	7e	39	17	13	60	78	a7	bb	52	4d	..\...~9...	..x..RM
00b0	20	c5	80	d7	fb	0e	c5	0b	de	6b	1e	44	e4	ce	47	2f		-k-D...G/
00c0	74	4c	74	a7	47	4f	fb	fc	fe	89	50	4a	03	3f	c5	fa	tLt.G0...	..PJ..?
00d0	dc	45	ff	36	0f	4a	2f	32	14	13	50	89	a4	2e	5c	01	..E-6-J/2...	..P...\.
00e0	f2	0e	7b	b7	99	70	3b	93	0d	f6	28	22	9d	ce	b4	28	..{...p;...	..("... (
00f0	a9	f6	bc	c1	45	8d	47	2c	ed	81	e4	ae	b9	fa	e5	7a	E-G,	z
0100	dd	b4	2c	f6	02	c5	01	31	4e	80	ea	d1	a3	31	37	d8	..,....1 N...	..17.
0110	a2	7b	75	6a	66	88	95	6e	d9	ff	97	84	4f	81	72	e3	..{ujf...n...	0..r.
0120	92	de	46	79	a2	11	cb	19	d0	0f	7c	0b	33	39	f6	6a	..Fy....	39-j
0130	0e	3c	19	43	da	a4	cd	e2	52	57	7e	5c	8f	92	4c	6b	..<-C....	RW...\..Lk

No.: 24898 · Time: 569.962624 · Source: 40.126.17.133 · Destination: 172.23.20.190 · Protocol: TCP · Length: 1514 · Info: [TCP Retransmission] 443 → 57242 [ACK] Seq=16986 Ack=7639 Win=4194560 Len=1460

4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation